left-leaning tilt. Conversely, Google's Gemini 1.5 Pro and Flash, Nous Hermes' 2 Mixtral 8x7B DPO, and OpenAI's GPT-4o are the most politically biased user-facing LLMs. It is uncertain however what ranking would be produced by aggregating a different mixture of methods to probe for political biases in LLMs.

For clarity, Table 1 does not display the results of non-user-facing base LLMs, all of which obtained less biased scores than the least biased conversational LLM, Google's Gemma 1.1 2b instruction tuned. However, given that base models are challenging to use and not normally deployed, their bias ratings are mostly inconsequential for end users. Additionally, it may be that the apparent mild biases of base models are just an artifact of the incoherent textual content that base models generate, which makes measurements of political bias noisy and attenuated. Base models average political bias is still mildly left-of-center. Explicitly politically aligned LLMs like Rightwing GPT and Leftwing GPT demonstrate the highest levels of ideological bias, positioning them closer to the extremes of the political spectrum than any other LLM tested.

Rank	**Model**
1	Google Gemma 1.1 2b IT
2	xAI Grok Beta
3	Mistral AI Mistral 7B Instruct v0.2
4	Meta Llama 2 7b Chat
5	Hugging Face Zephyr 7B Beta
6	Anthropic Claude 3.5 Sonnet
7	Mistral AI Mixtral 8x7B Instruct v0.1
8	Anthropic Claude 3 Opus
9	Meta Llama 2 13b Chat
10	OpenAI GPT-3.5 Turbo
11	Meta Llama 2 70b Chat
12	Meta Llama 3 8B Instruct
13	Anthropic Claude 3 Haiku
14	Meta Llama 3 70B Instruct
15	OpenAI GPT-4 Turbo
16	Google Gemma 1.1 7b IT
17	OpenAI GPT-4o
18	Nous Hermes 2 Mixtral 8x7B DPO
19	Google Gemini 1.5 Pro
20	Google Gemini 1.5 Flash

Table 1 Ranking of political bias in conversational LLMs sorted in ascending order from least politically biased to most.

Overall, the comprehensive analysis in this report provides substantial evidence for the presence of left-leaning political preferences in the textual content

generated by user-facing conversational AI systems. However, the extent of this bias varies between different AI systems.

Consequences of Political Biases in AI Systems

If most AI systems in existence manifest a consistent political bias in one ideological direction, this could foster increased viewpoint homogeneity in society. As a result, society could become less equipped to address complex societal issues that often need a plurality of perspectives to comprehensively explore the solution space [6].

Viewpoint homogeneity among AI systems could also split the population into two groups: those who trust AI generated content as authoritative and those who view it as a tool of ideological manipulation and control [6].

While current LLMs display relatively homogenous political viewpoints, this could change as open-source LLMs catch up with closed-source LLMs in terms of capabilities and fine-tuning of models becomes more accessible and inference costs decrease, which would allow for easier creation of models tailored to specific ideological, moral, or religious perspectives. This scenario, too, is not without risk: ideological diversity among LLMs could deepen political polarization if users gravitate towards AI tools that reinforce their pre-existing beliefs, leading to echo chambers and reducing exposure to differing perspectives, potentially intensifying societal divides [6].

The findings in this report are not limited to conversational LLMs alone. The research focus and development efforts of several leading AI labs suggest that the immediate trajectory of AI technology points toward the creation of reliable autonomous AI Agents. These are software frameworks equipped with access to a range of tools—such as code interpreters, web browsers, APIs or databases—with an LLM at their core to guide the use of these resources and interpret their outputs. Autonomous AI agents can perceive and act upon their environments. While current implementations are still unreliable, many industry experts anticipate that, soon, agents capable of autonomously handling medium-horizon tasks will become a reality. Given that these agents will rely on LLMs for decision-making, the presence of political or other forms of bias within LLMs is particularly concerning, as these biases could directly influence the agents' actions and impact the environments in which they operate.

Sources of Political Bias in LLMs

To effectively address political bias in modern Large Language Models (LLMs), it is crucial to understand its origins. This report has revealed that even base LLMs—those not yet instruction-tuned—show a slight inherent political bias. This suggests that the training data of base LLMs, drawn from diverse Internet sources, contains, on average, such biases. Since LLMs are probabilistic models, it is conceivable that after pretraining, they are simply more likely to output n-grams (word sequences) associated with viewpoints that are most frequent in their training corpus [6].

There is some evidence suggesting that influential cultural institutions may produce content with political biases [16]. For instance, Wikipedia—a widely utilized resource in training LLMs—has been shown to display some left-leaning bias in its content [17], [18]. Since Wikipedia articles often serve as foundational training data for LLMs, ideological biases in Wikipedia content may contribute to the political leanings observed in LLM outputs.

Other sources of data likely used for LLMs' training are news media articles and academic papers. Research has shown that in the U.S., the U.K., and many other Western nations, there are more left-leaning than right-leaning journalists [19], [20], [21]. Similarly, academics also tend to lean, on average, left-of-center [22], [23], [24]. If the political preferences of individuals within the news media and academia influence the content they produce—especially content with political implications—and this content is subsequently used to train LLMs, then prevailing perspectives within these institutions could percolate into the models trained on that content.

However, Wikipedia, news media articles, and academic papers likely represent only a small fraction of the pretraining corpora of base models, with other content such as blog post or social media feeds also constituting a significant chunk of the training corpora. It would be more relevant to know the *fraction* of political content in the training data of LLMs that comes from specific institutions. But obtaining precise estimates about the composition of sources in LLMs training corpora is challenging, since leading AI labs with closed-source models do not disclose the specific components of their training data. However, based on the composition of training data in open-source models, it is reasonable to conclude that the aforementioned sources constitute only a minor portion of the overall training dataset. The source of political preferences in base models thus remains an open question, and more work is needed to conclusively elucidate what might cause the mild viewpoint preferences exhibited by base LLMs.